Diego Sanchez Tirado

Stanford, CA | (984) 895-6386 | dsanchezt22@gmail.com | linkedin.com/in/diego-sanchez-tirado | github.com/dsanchezt22

EDUCATION

Stanford University Sept. 2025 – May 2029 (expected)
B.S. Electrical Engineering *Stanford, CA*
Coursework: Signals & Systems I (EE 102A), EE Fundamentals (ENGR 40M), AI Agents (CS 329Z), CS 106B/109, CME 100/102.

EXPERIENCE

Amazon Lab126 June 2026 – Sept. 2026
Hardware Development Engineer Intern | RAVEN / Governed Knowledge Engine *Sunnyvale, CA*
- Delivered an engineering-knowledge engine adopted across 3 teams, ingesting 1,500+ items daily from Outlook, Slack, and Confluence at $<5\%$ LLM-as-judge classification error; migrated retrieval from a Karpathy-style knowledge base to a gated S3 vector RAG store.
- Designed prompt-injection and PII gates before RAG indexing; raised leak-detection recall from 0.78 to 0.95 at $<3\%$ false positives, with F1 above 96% across every gate on a sealed adversarial benchmark spanning 10 leak classes.

Amazon Lab126 — Product Integrity & Reliability Lab June 2026 – Sept. 2026
Robotics Apprentice | Dexmate bimanual humanoid *Sunnyvale, CA*
- Designed and fabricated a ball-joint end-effector fixture that reduced manipulation-test reconfiguration to one adjustment; trained and validated ACT and diffusion imitation policies for box pickup across varied geometry, then extended them with residual RL.

Duna Strategies Apr. 2026 – Present
Founder & Lead Engineer *Remote*
- Built FStea AI for the Chinese beverage market: nine collectors spanning social, marketplace, retail-launch, search, and seasonality signals feed an engine that clusters, scores, and ranks trends into product concepts and generates full product specs.

Code Four (YC X25) July 2025 – Oct. 2025
GTM Team Lead *San Francisco, CA*
- Automated outreach to 10,000+ police contacts using AI agents and scrapers; led intern campaigns across 1,100+ departments, securing 70+ qualified meetings, 10+ signed MOUs, and 520+ officers onboarded in four months of nationwide outreach to agencies.

RESEARCH

Stanford Robust Systems Group — Prof. Subhasish Mitra Jan. 2026 – Present
Undergraduate Researcher | Area-Scalable Memory for LLM Inference (with Sam Dayo) *Stanford, CA*
- Modeled a $127\times$ throughput advantage for area-scalable mesh memory over 4-stack HBM4 at 2.1M-token context (3,708 vs. 29 tokens/s).
- Modeled LLM-inference throughput, latency, and memory footprint across on-chip memory organizations; related capacity, bandwidth density, and mesh/NoC topology to batch size and context length to identify bandwidth-bound regimes at long context.
- Benchmarked mesh-distributed memory against HBM under tensor/pipeline parallelism; produced compute, memory, and interconnect breakdowns identifying where long-context KV-cache traffic saturates each backend and where compute dominates instead.

Duke University — Dr. Charles Parker, Prof. Arturo F. Ehuan June 2023 – July 2025
Research Assistant | Nanomaterials & Thin Films Lab; Cyber Threat Intelligence *Durham, NC*
- Characterized 1–5 nm ALD Al_2O_3 coatings on CNT field emission: swept current vs. field at 10^{-6}–10^{-8} Torr, fit Fowler–Nordheim curves normalized by SEM emission area, and isolated a $\sim$50 V turn-on shift with no added degradation across thicknesses.
- Second author, *LATAM Financial Threat Landscape 2025* with Recorded Future; mapped five threat actors to MITRE ATT&CK via a Git-tracked IOC/TTP correlation pipeline; presented findings to financial-sector CISOs in Mexico City.

PROJECTS

Custom Electric Bike Conversion — *Independent* 2026
- Converted a mountain bike with a spot-welded 13S4P ($\sim$48 V, $\sim$0.8 kWh) Molicel P42A pack, copper-nickel strip, 80 A BMS, and Bafang BBSHD mid-drive; charged, balanced, and tested the pack under load before putting it into daily commuting service to Amazon.

4-Bit Adder/Subtractor with BCD Readout — *Digital Logic, NCSSM* 2024
- Designed and built a 4-bit adder/subtractor at the gate level: a ripple-carry chain of full adders composed from half-adders, two's-complement subtraction via XOR-inverted operand with injected carry, and a binary-to-BCD converter driving two 7-segment displays.

ErgoTouch — Split Ergonomic Keyboard — *Bowman Brockman Grant, $500* 2023 – 2024
- Designed a split, tented keyboard for arthritis and carpal-tunnel users: CAD resin-printed case, custom KiCad PCB, hand-soldered switch matrix, and RP2040 firmware with switches chosen by actuation force and travel; fourth keyboard built, first fully custom design.

Piezoelectric Typing-Force Energy Harvester — *NC Science & Engineering Fair* 2024
- Built a piezo rig (force plate, bridge rectifier, capacitor); swept force (2.5 N $\pm$ 1 N) and diameter (10/41 mm) over 360 actuations.

SKILLS & AWARDS

Hardware: Circuit design, KiCad PCB layout, RP2040 firmware, gate-level logic, soldering, SEM, high vacuum, CNC, 3D printing.

Architecture: HBM, NoC, TP/PP performance modeling, KV-cache analysis, batch and context-length scaling, gate-level digital logic.

Programming & AI: Python, C++, MATLAB, Git; RAG architecture, agent orchestration, Claude Code, Codex; fluent English/Spanish.

Awards: Congressional App Challenge 1st; Intel AI Global Impact Finalist; Bowman Brockman Grants ($2,500); FTC 5th/390.